Online Resource 1

Article: Reinforcement Learning for Investigating, Acting and Escalating in IT Incident Triage and Response: A Systematic Literature Review
Journal: Artificial Intelligence Review
Author: Charis Opol
Corresponding author: Charis Opol, Department of Computing and Technology, Faculty of Engineering, Design and Technology, Uganda Christian University, [**city**], Uganda
E-mail: B29654@students.ucu.ac.ug; charisopol@gmail.com

Contents. Table S1: outcome of every report sought for retrieval. Table S2: task, formulation, environment and main result of each included study. Table S3: coding of the review-question elements for each included study. Table S4: quality rating of each included study. These tables were Appendices A to D of the manuscript; their rows are unchanged. The decisions for all 131 screened records are in Online Resource 2.

Table S1 Outcome of every report sought for retrieval, with its identifier in the screening record (ID), title, publication year, priority group from title and abstract screening (Prio., 1 or 2) and outcome

ID	Report	Year	Prio.	Outcome
S006	A Hybrid Mathematical Framework for Dynamic Incident Prioritization Using Fuzzy Q-Learning and Text Analytics (Peralta et al. 2025)	2025	1	Included
S007	A leakage-free benchmark of offline policy learning for SOC alert triage (Okafor 2026)	2026	1	Included
S015	Adaptive Network Security Policies via Belief Aggregation and Rollout (Hammar et al. 2026)	2026	1	Included
S016	Adaptive NeuroDefense: Graph Neural Network and Reinforcement Learning-based Self-Healing Cybersecurity Architecture (Manoj Kumar et al. 2026)	2026	2	Excluded. EC3: evaluated only as a detector (accuracy on NSL-KDD, CICIDS-2017, UNSW-NB15); the DQN mitigation choices are not evaluated
S027	AI-Powered Incident Response Framework for Secure DevOps Environments (Bompally 2026)	2026	2	Excluded. EC3: the MDP action set is not specified and the agent is evaluated only with classification metrics (accuracy, AUC)
S031	An AI-Driven Cybersecurity Framework for IoT: Integrating LSTM-Based Anomaly Detection, Reinforcement Learning, and Post-Quantum Encryption (Saeed 2025)	2025	2	Included
S033	An Intelligent Proactive Cyber Threat Prediction Model Leveraging Self-Supervised Deep Learning and Reinforcement-Based Control Strategies (Noorul Hassan and Sivaraman 2026)	2026	2	Included
S038	Autonomous AI Decision Control Tower: A Deterministic Multi-Agent Decision Intelligence Framework for Enterprise NoOps (Singh 2026)	2026	1	Included
S039	Autonomous and Adaptive Cyber Incident Detection and Response in Industrial Cyber-Physical Systems Using Hierarchical Reinforcement Learning (Babar et al. 2026)	2026	2	Included
S043	CauseRL: Reinforcement Learning-based Random Walks for Root Cause Analysis in Microservices (Rojas Rodriguez et al. 2026)	2026	1	Included
S046	Combating alert fatigue with AlertPro: Context-aware alert prioritization using reinforcement learning for multi-step attack detection	2024	1	Not retrieved. Closed access: only the publisher landing page (abstract and section snippets) was obtained
S053	DAMS: Double Deep Q-Network Approach for DDoS Attack Mitigation in SDN (Khozam et al. 2025)	2025	2	Included
S056	Designing a Scalable and Adaptive Cyber Incident Response (DACIR) Framework for SMEs (Koupaei and Demirkol 2026)	2026	1	Included
S058	Digital Sovereignty by Design: Reinforcement Learning–Based Self-Healing for Cyber Resilience	2026	2	Not retrieved. Closed access (recorded at retrieval)
S059	Dispatching advanced and adaptive intrusion responses for IIoT-based systems (Elicha and Lopez 2026)	2026	1	Included
S060	Dual-Threshold Conformal Deferral for Trustworthy Security Alert Triage (Şahin and Mert 2026)	2026	1	Included

Table S1 (continued)

ID	Report	Year	Prio.	Outcome
S065	Explainable Reinforcement Learning for Automated Incident Response in Critical Infrastructure (Alauthman et al. 2026)	2026	1	Excluded. IC4: no quantitative evaluation. The paper specifies an evaluation plan and reports "indicative findings" as arrows, with no measured values
S066	Finding needles in a moving haystack: Prioritizing alerts with adversarial reinforcement learning (Tong et al. 2020)	2020	1	Included
S068	From Rules to Autonomy: Leveraging Reinforcement Learning for Real-Time Cyber Threat Mitigation	2026	2	Not retrieved. Closed access (recorded at retrieval)
S072	Hybrid LSTM-QEQ Framework for Enhanced Cyber Threat Detection and Response in Critical Power Grid Infrastructure	2026	2	Excluded. Out of scope (manual full-text judgement)
S073	Hybrid Reinforcement Learning and Agentic Artificial Intelligence Framework for Predictive Platform Operations and DevOps Orchestration (Nimmathi 2026)	2026	2	Included
S080	Large Language Models are Autonomous Cyber Defenders (Castro et al. 2025)	2025	2	Included
S081	Learning to Communicate in Multi-Agent Reinforcement Learning for Autonomous Cyber Defence (Contractor et al. 2025)	2025	2	Included
S084	Leveraging reinforcement learning for an efficient windows registry analysis during cyber incident response (Ghanem et al. 2026)	2026	1	Included
S085	LLM-Powered Deep Reinforcement Learning for Predictive Cyber Threat Mitigation (Varsha et al. 2025)	2025	2	Included
S087	Mission-Aware Cyber Incident Response Generation Using Reinforcement Learning (Jaber et al. 2025)	2025	2	Included
S088	MRCA: Metric-level Root Cause Analysis for Microservices via Multi-Modal Data (Wang et al. 2024)	2024	1	Included
S089	Multi-Agent Proximal Policy Optimization Framework for Autonomous Reliability Engineering in Kubernetes-Based Cloud-Native Distributed Systems (Aenugu 2026)	2026	2	Included
S093	Off-policy actor-critic deep reinforcement learning methods for alert prioritization in intrusion detection systems	2024	1	Not retrieved. Closed access (recorded at retrieval)
S094	Online Incident Response Planning under Model Misspecification through Bayesian Learning and Belief Quantization (Hammar and Li 2025)	2025	1	Included
S097	Optimizing cybersecurity incident response decisions using deep reinforcement learning (Alturkistani and El-Affendi 2022)	2022	2	Included
S102	PhishRL: An End-to-End Reinforcement Learning Pipeline for Automated Email Threat Triage (Bhosale et al. 2026)	2026	1	Included
S103	Policy-Based Profiles for Network Intrusion Response Systems (Hughes et al. 2022)	2022	1	Included
S104	Reinforced Autonomous Agents with Attack-Defense Exercises in Realistic Environments	2021	2	Excluded. IC5: full text is not in English (French)
S106	Reinforcement learning for an efficient and effective malware investigation during cyber incident response (Dunsin et al. 2025)	2025	1	Included
S108	Reinforcement Learning with AI for Autonomous Incident Response in DevSecOps: Using RL Agents to Detect, Classify, and Mitigate Security Threats in Cloud-Native DevOps Environments	2026	2	Not retrieved. No full text on file and no reason recorded
S109	Reinforcement Learning-Based Alert Prioritisation in Security Operation Centre: A Framework for Enhancing Cybersecurity in the Digital Economy	2023	1	Not retrieved. Closed access (recorded at retrieval)
S110	Research on Cross-Layer Link Fault Root Cause Localization Algorithm for Microservice Architecture Integrating Deep Reinforcement Learning (Wu et al. 2026)	2026	1	Included
S113	Risk-Aware SOC Alert Handling in Adaptive Cyber Defense with Reinforcement Learning	2026	1	Not retrieved. Closed access (recorded at retrieval)
S114	Risk-sensitive decision-making in security operations center alert escalation under resource constraints (Kovtun et al. 2026)	2026	1	Included
S116	Root Cause Analysis for Microservice Systems via Hierarchical Reinforcement Learning from Human Feedback (Wang et al. 2023)	2023	1	Included
S118	SAFE-HGX: A Hierarchical and Safety-Aware Reinforcement Learning Framework for Intrusion Response in Industrial Control Systems (Ravi et al. 2025)	2025	1	Included

Table S1 (continued)

ID	Report	Year	Prio.	Outcome
S119	Self-Healing and Intelligent Root Cause Analysis System for Data Link Faults in Cloud-Native Environments (Yuan 2026)	2026	2	Included
S123	Structural Generalization in Autonomous Cyber Incident Response with Message-Passing Neural Networks and Reinforcement Learning (Nyberg and Johnson 2024)	2024	2	Included
S125	TraceDiag: Adaptive, Interpretable, and Efficient Root Cause Analysis on Large-Scale Microservice Systems (Ding et al. 2023)	2023	1	Included
S130	XAI for Automated Prediction Model for Cyber Incident Response and Security Support in Cloud (Reddy Avula 2026)	2026	2	Excluded. EC3: response actions are described, but the model is evaluated only as a classifier of 26 memory dumps (accuracy, AUC)
S131	XAI-HDRL: An Intelligent Framework for Cyber Threat Prediction and Automated Security Response (Raj et al. 2026)	2026	2	Included

Table S2 Task, formulation and method, environment or data, and main reported result of each included study

Study	Year	Task	Formulation; method	Environment or data	Main reported result
Ding et al. (2023)	2023	Pruning service dependency graphs before root cause analysis	MDP; RL selecting from an engineer-derived pool of pruning actions	Microsoft Exchange traces over 345 days	Higher RCA accuracy and efficiency than baselines; integrated into Microsoft 365 Exchange
Wang et al. (2023)	2023	Building dependency graphs for RCA with engineer feedback	Hierarchical MDP; Hierarchical RL from human feedback	Pymicro and SynExchange synthetic data; Microsoft Exchange production trace	Human queries reduced from O(N^2) to $O(1)$ with higher RCA accuracy than baselines; deployed at Microsoft
Wang et al. (2024)	2024	Metric-level root cause analysis in microservices	MDP; Q-learning decides when to stop expanding a causal graph	Train Ticket and Online Boutique benchmark data	Higher accuracy than baselines at service and metric level and shorter causal-analysis time
Peralta et al. (2025)	2025	Prioritising enterprise IT incidents from report text	MDP over fuzzy states; Fuzzy Q-learning on BERT embeddings	10,000 incident reports from one enterprise (2,000 annotated) and a deployment trial	82.9% priority accuracy against 78.6% for BERT; overrides 15.2% to 6.2%, escalation rate 11.8% to 9.2% and MTTR down 9.1% in a before-after deployment comparison
Aenugu (2026)	2026	Autonomous reliability engineering in Kubernetes	Decentralised POMDP; Multi-agent PPO with a centralised critic	Simulated testbed of 40 microservices on 12 nodes and a staging cluster	MTTR 58.7 s against 148.6 s (static) and 96.3 s (single-agent PPO); MTTR rose from 41.2 s to 79.4 s at peak load
Nimmathi (2026)	2026	Predictive platform operations and remediation	MDP; DQN for discrete actions with PPO for resource allocation, plus coordinating agents	Simulated 500-node hybrid cloud over 90 days	MTTR 42.6 min to 29.1 min; manual interventions down 47.8%
Rojas Rodriguez et al. (2026)	2026	Ranking root-cause candidates in microservices	MDP with partial graph observation; PPO random walk over a causal graph	RCAEval benchmark (eight synthetic and microservice datasets)	Outperformed the best baselines on five of eight datasets
Singh (2026)	2026	Enterprise remediation decisions in cloud operations	Constrained MDP; Multi-agent scoring with PPO-tuned weights; IPO constraint solver	Google Cluster Trace v3 workloads on a prototype	MTTR 142 s against 198 s for AIOpsLab; GCR 99.9%; removing the deterministic guard left MTTR almost unchanged (144 s) but cut GCR to 72.3%
Wu et al. (2026)	2026	Cross-layer root cause localisation in microservices	MDP with a stop action; Attention-DQN	Kubernetes e-commerce system of 20 microservices with Chaos Mesh (10,000 faults)	Accuracy 96.5% with 3.2 average investigation steps against 6.1 for plain DQN
Yuan (2026)	2026	Self-healing for data-link faults in cloud-native systems	MDP; DQN	Kubernetes cluster with six injected fault types (2,000 samples)	Self-healing success improved by 31.9-90.8% over fixed scripts across fault types
Tong et al. (2020)	2020	Choosing which alerts to investigate under an analyst budget	Stochastic game; Adversarial deep RL in a double-oracle loop	Suricata alerts on CICIDS2017 and a credit-card fraud dataset	Outperformed baseline prioritisation policies, including when the adversary adapts
Alturkistani and El-Affendi (2022)	2022	Post-alert decisions in a SIEM	MDP; DQN	NSL-KDD and a phishing URL dataset replayed as alert streams	Accuracy 98.26% and 97.86% after 12 episodes
Bhosale et al. (2026)	2026	Triage of phishing alerts	Single-step MDP; DQN	150 simulated emails scored with VirusTotal and AlienVault OTX	Accuracy 89.3% against 82.0%; medium-risk F1 0.839 against 0.725; 44% of emails escalated
Kovtun et al. (2026)	2026	Risk-sensitive alert escalation under analyst capacity limits	Constrained risk-sensitive POMDP with belief; PPO with an entropic criterion and KL regularisation to a playbook policy	Alert arrival processes from CIC-IDS2017 and UNSW-NB15 with a simulated eight-analyst queue	Deferring cut false escalations from 7.2% to 5.0% and 9.4% to 6.6%; the miss and budget constraints held in 14 of 15 and 13 of 15 seeds

Table S2 (continued)

Study	Year	Task	Formulation; method	Environment or data	Main reported result
Okafor (2026)	2026	SOC alert triage learned offline from reconstructed trajectories	Offline MDP (pseudo-trajectories); BC, recurrent BC, GRU/LSTM, CQL, IQL, Decision Transformer	UNSW-NB15 and CIC-IDS-2017 flows regrouped into decision episodes	Recurrent behavioural cloning best (MCC 0.752 and 0.621); CQL, IQL and Decision Transformer degenerated into trivial detectors that F1 and ROC-AUC concealed and MCC revealed
Şahin and Mert (2026)	2026	Alert investigation with certified deferral to analysts	Episodic MDP per alert; PPO investigation agent with a dual-threshold conformal deferral layer	Simulated SOC with labels drawn from CIC-IDS2017, CSE-CIC-IDS2018 and UNSW-NB15, plus four classifiers on real flows	Automated 73.7% of decisions at budgets of 0.01 with realised rates 0.0099 and 0.0101, deferring about 26%; language models failed as a second-stage investigator
Nyberg and Johnson (2024)	2024	Incident response that generalises across network structures	MDP over a relational graph; PPO with a message-passing neural network	CAGE-2 with seven network variants	Graph agents generalised to unseen networks without retraining but scored below MLP agents trained on each variant
Castro et al. (2025)	2025	Multi-agent cyber defence with RL and LLM agents	Multi-agent partially observable game (CAGE-4); KEEP graph PPO agents and LLM agents	CybORG CAGE-4	All-RL team mean reward -493 (sd 95.9) against -2547.2 (sd 498.8) for all-LLM; LLM agents misread communication vectors
Contractor et al. (2025)	2025	Cooperative cyber defence with learned communication	Multi-agent partially observable MDP; DIAL against QMIX	CybORG with three network configurations	With 50% detection, unmasking and blocking, DIAL agents returned -26.4 against -43.4 for QMIX on the large network
Hammar and Li (2025)	2025	Online incident response planning under model misspecification	POMDP with belief (particle filter); Bayesian model learning with belief quantisation and dynamic programming	CAGE-2 benchmark	Under misspecification cost 35.91 against 91.51 or more for all other methods; slightly worse than the best methods without misspecification
Jaber et al. (2025)	2025	Mission-aware incident response on a military network	MDP; Q-learning with softmax policy	Cyber Security Learning Environment digital twin	78.4% mission success rate
Khozam et al. (2025)	2025	DDoS mitigation in a software-defined network	MDP; Double DQN	Mininet emulation with generated benign and attack traffic	Attacker bandwidth reduced by 96.7%; benign share of delivered packets rose from 29.6% to 85.8% (ICMP) but only to 39.4% under SYN floods
Varsha et al. (2025)	2025	Proactive cyber defence against an adaptive attacker	MDP; Dueling DQN guided by an LLM	Custom red-blue simulator over 15 ATT&CK techniques	80% win rate over five trials against 26.7% for the heuristic
Hammar et al. (2026)	2026	Adapting intrusion recovery and response policies	POMDP with belief (particle filter); Feature-based belief aggregation and rollout	Physical testbed with emulated attacks (24,386 measurements) and the CAGE-2 benchmark	On par with the state of the art on CAGE-2; adapted to system change in about 15-19 s against more than 30 min for PPO; PPO variance about ten times higher
Koupaei and Demirkol (2026)	2026	Cyber incident response for small and medium enterprises	POMDP with belief (Dempster-Shafer); One-step lookahead with an uncertainty penalty and post-event parameter learning	Simulated 50-asset network with a four-phase ransomware scenario	MTTR 58.7 min against 95.3 min for the static baseline; expected loss USD 39,500 against 84,000
Noorul Hassan and Sivaraman (2026)	2026	Proactive threat prediction with RL-controlled mitigation	MDP; DQN on self-supervised threat embeddings	CIC-IDS2017 and UNSW-NB15 (RL environment not described)	Reports containment probability 0.954 and 82 ms response latency
Raj et al. (2026)	2026	Threat prediction and automated security response	MDP; PPO response module in a hybrid pipeline	CICIDS2017 in an NS-3 simulation	Attack mitigation rate 97.52% and response time 12.47 ms
Hughes et al. (2022)	2022	Intrusion response aligned with organisational response policies	MDP; PPO with policy-specific reward profiles	Mininet network integrated with a Tennessee Eastman process in Matlab	Aggressive profile converged on blocking all traffic; balanced profile on disabling the web server interface

Table S2 (continued)

Study	Year	Task	Formulation; method	Environment or data	Main reported result
Ravi et al. (2025)	2025	Safety-aware intrusion response in ICS	Hierarchical multi-agent MDP with recurrent state; Offline-to-online hierarchical MARL (PPO/SAC) with a policy shield	SafeHGXEnv ICS simulator with Sandworm, Wizard Spider and Turla profiles	526-623 shield interventions per scenario; 9-11 times more false actions and 74-77% higher cost than PPO; recall 0.96-0.99 against 0-0.084 for PPO
Saeed (2025)	2025	Automated threat response in an IoT network	MDP; Tabular Q-learning	Simulated IoT network (RL environment not described)	Learned policy blocks mild attacks and counter-attacks severe ones; reported only as a three-state Q-table
Babar et al. (2026)	2026	Adapting detection and response thresholds in industrial CPS	Hierarchical MDP; Four hierarchical RL variants (h-DQN, option-critic)	Simulated industrial CPS with a network data generator	All four hierarchical variants outperformed the static threshold method
Elicha and Lopez (2026)	2026	Early automated intrusion response for IIoT	MDP; Expert system plus neural fitted Q-iteration	Industrial testbed emulating energy generation	Mean time to response about 4 min (SSH brute force) and 0.8 min (malware download); recovery took about 7 min when the attacker countered the response
Dunsin et al. (2025)	2025	Malware investigation during incident response	MDP; Tabular Q-learning	Gym environments built from infected Windows virtual machines	94% agreement with ideal actions and shorter investigation time than a human expert
Ghanem et al. (2026)	2026	Prioritising Windows Registry analysis during incident response	MDP with expert-set transitions; Dynamic programming with bounded tabular RL refinement and rule-based AI	Four forensic datasets across Windows versions and scenarios	Investigation time cut by up to 68% and adjudicated relevant artefacts up by up to 35% against analyst-assisted baselines (10 seeded runs)

A dash means the element is absent or not reported.

Table S3 Coding of the review-question elements for each included study: information gathering, response action or disposition, escalation, constraint (with the handling of a blocked action) and robustness test

Study	Information gathering	Response action or disposition	Escalation	Constraint (and blocked-action handling)	Robustness test
Ding et al. (2023)	Decide which components to exclude from analysis	–	–	Action masking: Interpretable pruning action pool derived from engineers	–
Wang et al. (2023)	Choose which dependency to ask engineers about	–	–	None	–
Wang et al. (2024)	Expand a node or stop expanding the causal graph	–	–	Action masking: Expanded nodes masked	–
Peralta et al. (2025)	–	Raise, keep or lower priority	–	None: Fuzzy rule base defines states; no enforcement	Synthetic terminology drift: accuracy 82.9% to 79.4%
Aenugu (2026)	–	Scale up or down, restart pod, reroute traffic, no-op	–	None	Four workload levels
Nimmathi (2026)	–	Restart, reroute, rollback, scale, failover	–	None: Plans checked against predefined safety constraints (not evaluated)	–
Rojas Rodriguez et al. (2026)	Choose which node of the causal graph to visit next	–	–	Action masking: Invalid-action masking	Held-out cases on eight datasets
Singh (2026)	–	Scale, throttle, restart, isolate, rollback	Hand-off to a human when the safety check fails (rule-triggered)	Hard filter or shield: Feasible action region plus deterministic execution guard; CMDP cost constraints. Blocked or uncertain cases: Routed to a human	–
Wu et al. (2026)	Move to a neighbouring node or stop and declare the root cause	–	–	Action masking: Pruning of low-probability nodes	Dynamic topology (accuracy above 90%); fault severity levels
Yuan (2026)	–	Scale, degrade service, shift traffic, rebuild pods, restart sidecars	–	Reward penalty: Penalty of -2 for destructive actions	–
Tong et al. (2020)	Select the subset of alerts for analysts to investigate within a budget	–	–	Expected-cost constraint: Hard investigation budget per period	Adversary adapting to the policy
Alturkistani and El-Affendi (2022)	–	Fix	Escalate to the next tier of SOC operators	None	–
Bhosale et al. (2026)	–	Allow or monitor	Escalate (creates a ServiceNow incident)	Reward penalty: Asymmetric reward penalising under-escalation	–
Kovtun et al. (2026)	Defer to gather more context	Shallow triage	Escalate to deep analysis	Expected-cost constraint: Bounds on deep-analysis use and on the attack-miss rate; deep action unavailable when the analysis resource is unavailable	Load near capacity; chronological test split; 15 seeds
Okafor (2026)	Investigate action	Contain action; Dismiss	Containment left to analysts in the deployment design (not learned)	Hard filter or shield: Deterministic severity floor in deployment: medium and high alerts cannot be auto-dismissed (design, not evaluated)	Chronological future test fold; five seeds; two datasets

Table S3 (continued)

Study	Information gathering	Response action or disposition	Escalation	Constraint (and blocked-action handling)	Robustness test
Şahin and Mert (2026)	Query logs, investigate lateral movement, query threat intelligence	Close as false positive	Escalate to incident response; defer the uncertain band to an analyst	Expected-cost constraint: Statistical error budgets on benign escalation and threat auto-close. Blocked or uncertain cases: Deferred to an analyst	Hardened regime; cross-dataset shift (deferral rose to 79-99% for the worst classifiers)
Nyberg and Johnson (2024)	Analyse	Remove, restore	–	Reward penalty: Restore actions are penalised	Unseen network sizes and an unseen attacker policy
Castro et al. (2025)	Analyse and monitor	Remove, restore, deploy decoy, block or allow zone traffic	–	Reward penalty: Penalties when legitimate users lose service	Five attacker strategies
Contractor et al. (2025)	Analyse (unmasked on detection or a message)	Remove, restore, block	–	Action masking: Restore unmasked only for persistent threats; analyse only after a detection or message	Detection rate lowered to 50%; false alerts from benign users; larger networks
Hammar and Li (2025)	Analyse (CAGE-2)	Decoys, malware removal, restore	–	Reward penalty: Service disruption and compromise costs	Misspecified model of decoy effectiveness
Jaber et al. (2025)	Increase logging	Kill process, block outbound, filter traffic, take down network	–	Reward penalty: Mission value threshold and risk penalty in the reward	–
Khozam et al. (2025)	–	Raise or lower a host's bandwidth, or do nothing	–	Reward penalty: Reward on benign packet loss and delay	Three attack types and randomised host positions
Varsha et al. (2025)	–	Deploy defences across ATT&CK techniques	–	Expected-cost constraint: Defence budget of 100 units, replenished each round	–
Hammar et al. (2026)	Analyse control (CAGE-2)	Recovery, decoy, removal and restore controls	–	Reward penalty: Recovery and service disruption carry costs	Non-stationary scenarios on the testbed and CAGE-2
Koupaei and Demirkol (2026)	–	Micro-playbooks (isolate host, revoke credentials, restore, preserve evidence)	Notify a managed security provider (playbook)	Hard filter or shield: Allowable action set restricted by cost and time limits	–
Noorul Hassan and Sivaraman (2026)	–	Firewall rules, throttling, isolation, access revocation	–	Reward penalty: Mitigation cost in the reward	–
Raj et al. (2026)	–	Mitigation actions (not listed)	–	None	–
Hughes et al. (2022)	–	26 actions including block, shut down, disable interface, adjust plant valves, no action	–	Reward penalty: Aggressive, balanced and remediation profiles encoded in the reward	–
Ravi et al. (2025)	–	Deception, containment and isolation actions	–	Hard filter or shield: Rule-based policy shield (legal, SLA and safety rules). Blocked or uncertain cases: Replaced by a safer alternative or a no-op, with a reward penalty	Trained on Sandworm and tested on unseen Wizard Spider and Turla profiles
Saeed (2025)	–	Block or counter-attack	–	None	–
Babar et al. (2026)	–	Select detection and response threshold ranges	–	Reward penalty: Cost of changing the selected range	–
Elicha and Lopez (2026)	–	Void, restrict traffic, restrict credentials, kill processes	–	Hard filter or shield: Expert rules decide known threats first; action reputation register; void action to ignore false positives	–
Dunsin et al. (2025)	Choose the next forensic analysis step	–	–	None	–

Table S3 (continued)

Study	Information gathering	Response action or disposition	Escalation	Constraint (and blocked-action handling)	Robustness test
Ghanem et al. (2026)	Choose the next registry or timeline analysis step	–	–	Action masking: State-dependent action sets; expert policy graphs	–

Items Q1–Q8 are defined in Table 7 of the article. “Items met” counts only the items rated Yes; items rated Part are not counted.

Table S4 Quality rating of each included study on checklist items Q1–Q8 (Yes, met; Part, partly met; No, not met) and the number of items met

Study	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Items met
Ding et al. (2023)	Yes	Yes	Yes	Yes	No	Part	Yes	Part	5
Wang et al. (2023)	Yes	Yes	Yes	Yes	Part	Part	No	Part	4
Wang et al. (2024)	Yes	Yes	Yes	Yes	No	Part	Yes	Part	5
Peralta et al. (2025)	Yes	Part	Yes	Yes	Yes	Yes	Yes	Part	6
Aenugu (2026)	Part	Part	Yes	Part	Part	Part	Part	No	1
Nimmathi (2026)	Yes	Part	Yes	Part	No	No	No	Part	2
Rojas Rodriguez et al. (2026)	Yes	Yes	Yes	Yes	Part	Yes	Yes	Part	6
Singh (2026)	Part	Part	Yes	Part	No	No	Part	Part	1
Wu et al. (2026)	Yes	Yes	Yes	Yes	No	Part	Yes	Part	5
Yuan (2026)	Yes	Part	Yes	Yes	Part	Part	Yes	Part	4
Tong et al. (2020)	Yes	Yes	Yes	Yes	No	Part	No	No	4
Alturkistani and El-Affendi (2022)	Part	Yes	No	Part	No	No	No	Part	1
Bhosale et al. (2026)	Yes	Part	Yes	Yes	No	No	Yes	Part	4
Kovtun et al. (2026)	Yes	Yes	Part	Yes	Yes	Yes	Yes	Part	6
Okafor (2026)	Yes	Yes	Part	Yes	Yes	Yes	Yes	Part	6
Şahin and Mert (2026)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8
Nyberg and Johnson (2024)	Yes	Yes	Part	Yes	Yes	Yes	Yes	Yes	7
Castro et al. (2025)	Yes	Yes	Yes	Part	Part	Part	Yes	Yes	5
Contractor et al. (2025)	Yes	Yes	Part	Yes	Yes	No	No	Part	4
Hammar and Li (2025)	Yes	Yes	Part	Yes	Yes	Yes	Yes	Part	6
Jaber et al. (2025)	Yes	Part	No	No	No	No	Part	No	1
Khozam et al. (2025)	Yes	Yes	No	Part	No	Part	Yes	Yes	4
Varsha et al. (2025)	Part	Yes	Yes	Yes	Yes	No	Yes	Part	5
Hammar et al. (2026)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8
Koupaei and Demirkol (2026)	Part	Part	Yes	Yes	Part	No	Yes	No	3
Noorul Hassan and Sivaraman (2026)	Part	No	No	No	No	No	No	No	0
Raj et al. (2026)	Part	Part	Part	Yes	No	No	No	No	1
Hughes et al. (2022)	Yes	Yes	No	Part	No	No	Part	No	2
Ravi et al. (2025)	Part	Part	Part	Yes	Yes	Yes	No	No	3
Saeed (2025)	Part	Part	No	No	No	No	Yes	No	1
Babar et al. (2026)	Yes	Part	Yes	Part	No	No	Part	No	2
Elicha and Lopez (2026)	Yes	Part	No	Part	Part	No	Part	No	1
Dunsin et al. (2025)	Yes	Part	Part	Part	No	No	Part	No	1
Ghanem et al. (2026)	Yes	Yes	Yes	Yes	Yes	Part	Yes	Part	6

References

Aenugu GR (2026) Multi-agent proximal policy optimization framework for autonomous reliability engineering in Kubernetes-based cloud-native distributed systems. In: Proceedings of the 7th International Conference on Computational Vision and Bio Inspired Computing (ICCVBIC 2026), pp 500–506, https://doi.org/10.1109/ICCVBIC71195.2026.11688912

Alauthman M, Almomani A, Al-Qerem A, et al (2026) Explainable reinforcement learning for automated incident response in critical infrastructure. In: Proceedings of

the 2nd International Conference on Computational Intelligence Approaches and Applications (ICCIAA 2026), pp 1–7, https://doi.org/10.1109/ICCIAA68481.2026.11543556

Alturkistani H, El-Affendi MA (2022) Optimizing cybersecurity incident response decisions using deep reinforcement learning. Int J Electr Comput Eng 12(6):6768–6776. https://doi.org/10.11591/ijece.v12i6.pp6768-6776

Babar A, Halabi T, Zulkernine M (2026) Autonomous and adaptive cyber incident detection and response in industrial cyber-physical systems using hierarchical reinforcement learning. ACM Trans Cyber-Phys Syst 10(1):8. https://doi.org/10.1145/3765622

Bhosale A, Rambhia J, Kharche S, et al (2026) PhishRL: an end-to-end reinforcement learning pipeline for automated email threat triage. In: Proceedings of the 2026 6th International Conference on Intelligent Technologies (CONIT 2026), https://doi.org/10.1109/CONIT69683.2026.11620809

Bompally SD (2026) AI-powered incident response framework for secure DevOps environments. In: Proceedings of the 2026 IEEE International Conference on Emerging Computing and Intelligent Technologies (ICoECIT 2026), https://doi.org/10.1109/ICoECIT68303.2026.11497511

Castro SR, Campbell R, Lau N, et al (2025) Large language models are autonomous cyber defenders. In: Proceedings of the 2025 IEEE Conference on Artificial Intelligence (CAI 2025), pp 1125–1132, https://doi.org/10.1109/CAI64502.2025.00195

Contractor F, Li L, Al Mallah R (2025) Learning to communicate in multi-agent reinforcement learning for autonomous cyber defence. In: Proceedings of the International Conference on Machine Learning and Cybernetics (ICMLC 2025), pp 26–31, https://doi.org/10.1109/ICMLC66258.2025.11280109

Ding R, et al (2023) TraceDiag: adaptive, interpretable, and efficient root cause analysis on large-scale microservice systems. In: Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023), pp 1762–1773, https://doi.org/10.1145/3611643.3613864

Dunsin D, Ghanem MC, Ouazzane K, et al (2025) Reinforcement learning for an efficient and effective malware investigation during cyber incident response. High-Confidence Computing 5(3):100299. https://doi.org/10.1016/j.hcc.2025.100299

Elicha J, Lopez J (2026) Dispatching advanced and adaptive intrusion responses for IIoT-based systems. Future Gener Comput Syst 179:108314. https://doi.org/10.1016/j.future.2025.108314

Ghanem MC, Wojtczak D, Benkhelifa E, et al (2026) Leveraging reinforcement learning for an efficient Windows registry analysis during cyber incident response. Sci Rep 16(1):26867. https://doi.org/10.1038/s41598-026-57787-6

Hammar K, Li T (2025) Online incident response planning under model misspecification through Bayesian learning and belief quantization. In: Proceedings of the 18th ACM Workshop on Artificial Intelligence and Security (AISec 2025), pp 40–51, https://doi.org/10.1145/3733799.3762965

Hammar K, Li Y, Alpcan T, et al (2026) Adaptive network security policies via belief aggregation and rollout. IEEE Trans Inf Forensics Secur 21:6544–6559. https://doi.org/10.1109/TIFS.2026.3709131

Hughes K, McLaughlin K, Sezer S (2022) Policy-based profiles for network intrusion response systems. In: Proceedings of the 2022 IEEE International Conference on Cyber Security and Resilience (CSR 2022), pp 279–286, https://doi.org/10.1109/CSR54599.2022.9850304

Jaber A, Endregard M, Mancini F, et al (2025) Mission-aware cyber incident response generation using reinforcement learning. In: Proceedings of the International Conference on Military Communication and Information Systems (ICMCIS 2025), https://doi.org/10.1109/ICMCIS64378.2025.11047892

Khozam S, Blanc G, Tixeuil S, et al (2025) DAMS: double deep Q-network approach for DDoS attack mitigation in SDN. In: ACM International Conference Proceeding Series, pp 1049–1056, https://doi.org/10.1145/3726122.3726282

Koupaei ANA, Demirkol I (2026) Designing a scalable and adaptive cyber incident response (DACIR) framework for SMEs. Journal of Intelligent Decision Making and Information Science 3(5S):131–141. https://doi.org/10.59543/jidmis.v3.1073

Kovtun V, Huralnyk A, Husak L, et al (2026) Risk-sensitive decision-making in security operations center alert escalation under resource constraints. In: CEUR Workshop Proceedings, pp 65–83

Manoj Kumar R, Srinivasan K, Rohidh S, et al (2026) Adaptive NeuroDefense: graph neural network and reinforcement learning-based self-healing cybersecurity architecture. In: Proceedings of the 4th International Conference on Augmented Intelligence and Sustainable Systems (ICAISS 2026), pp 251–257, https://doi.org/10.1109/ICAISS68683.2026.11525610

Nimmathi SR (2026) Hybrid reinforcement learning and agentic artificial intelligence framework for predictive platform operations and DevOps orchestration. In: Proceedings of the 7th International Conference on Computational Vision and Bio Inspired Computing (ICCVBIC 2026), pp 1197–1203, https://doi.org/10.1109/ICCVBIC71195.2026.11689187

Noorul Hassan S, Sivaraman K (2026) An intelligent proactive cyber threat prediction model leveraging self-supervised deep learning and reinforcement-based control strategies. In: Proceedings of the 2026 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT 2026), https://doi.org/10.1109/TQCEBT67648.2026.11681196

Nyberg J, Johnson P (2024) Structural generalization in autonomous cyber incident response with message-passing neural networks and reinforcement learning. In: Proceedings of the 2024 IEEE International Conference on Cyber Security and Resilience (CSR 2024), pp 282–289, https://doi.org/10.1109/CSR61664.2024.10679456

Okafor E (2026) A leakage-free benchmark of offline policy learning for SOC alert triage. Mach Learn Appl 25:100984. https://doi.org/10.1016/j.mlwa.2026.100984

Peralta A, Olivas JA, Navarro-Illana P, et al (2025) A hybrid mathematical framework for dynamic incident prioritization using fuzzy Q-learning and text analytics. Mathematics 13(12):1941. https://doi.org/10.3390/math13121941

Raj AP, Gantayat SS, Sharma KV (2026) XAI-HDRL: an intelligent framework for cyber threat prediction and automated security response. In: Proceedings of the 7th International Conference on Smart Systems and Inventive Technology (ICSSIT 2026), pp 1115–1123, https://doi.org/10.1109/ICSSIT69151.2026.11656470

Ravi RV, Priyanga KK, Adarsh MD (2025) SAFE-HGX: a hierarchical and safety-aware reinforcement learning framework for intrusion response in industrial control systems. In: Proceedings of the 2025 1st International Conference on Advancement in Futuristic Technologies (ICAFT 2025), https://doi.org/10.1109/ICAFT66710.2025.11453104

Reddy Avula CS (2026) XAI for automated prediction model for cyber incident response and security support in cloud. In: Proceedings of the 2026 2nd International Conference on Big Data and Machine Learning (ICBDML 2026), https://doi.org/10.1109/ICBDML68582.2026.11544635

Rojas Rodriguez JS, Lahmadi A, Rusinowitch M (2026) CauseRL: reinforcement learning-based random walks for root cause analysis in microservices. In: Proceedings of the ACM Symposium on Applied Computing (SAC 2026), pp 257–264, https://doi.org/10.1145/3748522.3779836

Saeed MM (2025) An AI-driven cybersecurity framework for IoT: integrating LSTM-based anomaly detection, reinforcement learning, and post-quantum encryption. IEEE Access 13:104027–104036. https://doi.org/10.1109/ACCESS.2025.3576506

Şahin F, Mert NS (2026) Dual-threshold conformal deferral for trustworthy security alert triage. Electronics 15(18):4084. https://doi.org/10.3390/electronics15184084

Singh T (2026) Autonomous AI decision control tower: a deterministic multi-agent decision intelligence framework for enterprise NoOps. In: Proceedings of the 2026 IEEE 1st International Conference on Artificial Intelligence Implementation and Applications (ICAIIA 2026), pp 190–195, https://doi.org/10.1109/ICAIIA70172.2026.11691884

Tong L, Laszka A, Yan C, et al (2020) Finding needles in a moving haystack: prioritizing alerts with adversarial reinforcement learning. In: Proceedings of the 34th AAAI Conference on Artificial Intelligence (AAAI 2020), pp 946–953

Varsha MP, Samyuktha S, Manochithra S, et al (2025) LLM-powered deep reinforcement learning for predictive cyber threat mitigation. In: Proceedings of the 5th International Conference on Ubiquitous Computing and Intelligent Information Systems (ICUIS 2025), pp 1020–1025, https://doi.org/10.1109/ICUIS67429.2025.11380737

Wang L, et al (2023) Root cause analysis for microservice systems via hierarchical reinforcement learning from human feedback. In: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2023), pp 5116–5125, https://doi.org/10.1145/3580305.3599934

Wang Y, Zhu Z, Fu Q, et al (2024) MRCA: metric-level root cause analysis for microservices via multi-modal data. In: Proceedings of the 39th ACM/IEEE International Conference on Automated Software Engineering (ASE 2024), pp 1057–1068, https://doi.org/10.1145/3691620.3695485

Wu Z, Fu H, Wang L, et al (2026) Research on cross-layer link fault root cause localization algorithm for microservice architecture integrating deep reinforcement learning. In: Proceedings of the 2026 IEEE 6th International Conference on Electronic Technology, Communication and Information (ICETCI 2026), pp 421–426, https://doi.org/10.1109/ICETCI68937.2026.11610700

Yuan Y (2026) Self-healing and intelligent root cause analysis system for data link faults in cloud-native environments. In: Proceedings of the 2026 6th International Conference on Applied Mathematics, Modelling and Intelligent Computing (CAMMIC 2026), pp 941–946, https://doi.org/10.1145/3817124.3817267